TOC-PRES-24-6032
Revision 0

Development of Double Shell Tank Annulus Vacuum Crawler System

Prepared for the U.S. Department of Energy
Assistant Secretary for Environmental Management

Contractor for the U.S. Department of Energy
Office of River Protection under Contract DE-AC27-08RV14800



P.O. Box 850
Richland, Washington 99352

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D. Reid
Washington River Protection Solutions

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STUDENT SUMMER INTERNSHIP TECHNICAL REPORT

Development of Double Shell Tank Annulus Vacuum Crawler System

DOE-FIU SCIENCE & TECHNOLOGY WORKFORCE DEVELOPMENT PROGRAM

Date submitted:

January,2025

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Submitted to:

U.S. Department of Energy
Office of Environmental Management
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FIU | Applied Research Center
FLORIDA INTERNATIONAL UNIVERSITY

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EXECUTIVE SUMMARY

This research work has been supported by the DOE-FIU Science & Technology Workforce Development Initiative, an innovative program developed by the U.S. Department of Energy's Office of Environmental Management (DOE-EM) and Florida International University's Applied Research Center (FIU-ARC). The initiative was supported by Sr. Research Scientist Dr. Amer Awwad. During the summer of 2024, DOE Fellow intern Pedro Chaviano undertook a 10-week internship at Hanford and Washington River Protection Solutions (WRPS) Chief Technology Office (CTO), under the supervision of Kayle Boomer, with guidance from Dr. Douglas J. Reid and Rogeiro G. Renteria.

The project, which began on May 28, 2024, and concluded on August 9, 2024, focused on the development of a robotic vacuum crawler aimed at enhancing inspection and cleaning operations within the annulus space of double-shell tanks (DSTs). The work was aligned with the needs of WRPS's Tank and Pipeline Integrity (TAPI) organization, as outlined in the Technology Element Description Summary (TEDS) MTW-89. This document highlights the necessity of remote cleaning for concrete surfaces painted with a special protective coating (SPC), required for RCRA permit compliance, to enable proper inspection and repair in high-radiation environments.

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TRADEMARKS

Exair Line Vac 6080 is a registered trademark of EXAIR, 11510 Goldcoast Dr Cincinnati, OH, USA.

Intel® RealSense™ D415 camera is a registered trademark of Intel Corporation, USA.

Stratasys F170 3D printer is a registered trademark of Stratasys, 7665 Commerce Way Eden Prairie, MN USA.

LIST OF ACRONYMS

ABS	Acrylonitrile butadiene styrene
VCS	Vacuum Crawler System
CTF	Cold Test Facility
CTO	Chief Technology Office
DOE-EM	Department of Energy's Office of Environmental Management
DSTs	Double-shell tanks
FIU	Florida International University
FIU-ARC	Florida International University's Applied Research Center
SPC	Special protective coating
TAPI	Tank and Pipeline Integrity
TEDS	Technology Element Description Summary
WRPS	Washington River Protection Solutions

1. INTRODUCTION

During the summer of 2024, Pedro Chaviano, a master's candidate in electrical engineering from Florida International University (FIU), joined the Chief Technology Office (CTO) at Washington River Protection Solutions (WRPS) to contribute to a key research and development project for the Tank and Pipeline Integrity (TAPI) organization, as outlined in the Technology Element Description Summary (TEDS) MTW-89, which highlights the need for remote cleaning solutions. This project focused on the design and development of a remotely operated robotic vacuum crawler intended for the inspection and cleaning of the annulus space in double-shell tanks (DSTs), Figure 1, at the DOE's Hanford site.

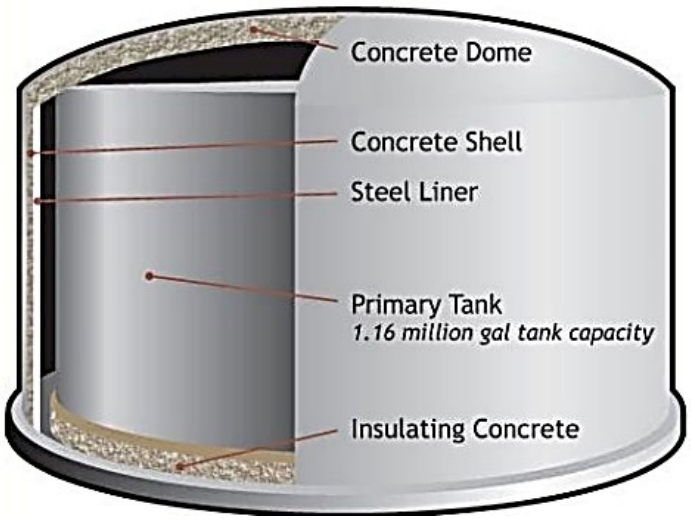


Figure 1. (28) In-Service Underground Double-Shell Tanks constructed in 1968-1986

1.1 Project Overview

Inspection of the double shell tanks include ultrasonic testing of the primary tank walls as well as the secondary tank bottoms to test for wall thinning. The primary tank wall must be corrosion free for the test to be conducted. Some of these tanks require extensive work to remove corrosion on the outer surface of the primary tank before ultrasonic testing could be performed. This results in significant amounts of debris on the annulus floor. This debris interferes with efforts to perform ultrasonic testing on the secondary tank bottom. The annulus space, approximately 30 inches wide with access points ranging from 12 to 24 inches in diameter, presents significant engineering challenges for robotic systems. Figure 1 depicts one of the 28 in-service DSTs constructed between 1968 and 1986. The tanks accumulate debris over time, with some tanks experiencing debris buildup up to 1 foot deep (Figure 2).



Figure 2. Debris Inside Double Shell Tank AY-101

The debris must be removed to enable accurate inspection of the wall thickness and to facilitate the repair and reinforcement of areas where degradation has occurred due to environmental contact with the outer wall, as shown in Figure 3.

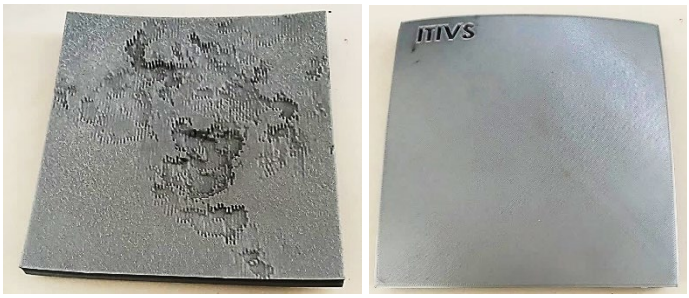


Figure 3. Representation of tank surface degradation on left and surface without degradation on right (*3D printed model)*

The vacuum crawler system (VCS), as outlined in section vii, includes various components essential for its operation. The crawler is equipped with two EXAIR Line Vac pneumatic vacuum modules mounted in series—one on the crawler itself and the other positioned higher up on the umbilical harness. This module, integrated into the system, maximizes pull force and airflow at the cleaning nozzle. The system features a durable design with components such as a heavy-duty in-line vacuum module, advanced control capabilities including real-time video feedback with low latency.

The crawler's robust architecture allows it to traverse obstacles and welds up to ½ inch in height, through utilizing differential drive methods. The system’s protype chassis is crafted from acrylonitrile butadiene styrene (ABS) 3D printed plastic and will be upgraded to stainless steel, anodized aluminum, or non-porous plastics, ensuring material compatibility with the nuclear environment and meeting stringent radiation tolerance requirements.

Key features of the design include a pneumatic vacuum system, operated by an appropriately sized air compressor rated at 185 scfm. The crawlers future state will include mechanical assemblies designed to agitate surfaces and reduce particle sizes to ½ inch or less, enhancing the vacuum efficiency and reducing airborne debris.

Pedro’s internship focused on addressing critical engineering challenges such as suction power, filtration, maneuverability, and deployment strategies. Testing at the Cold Test Facility (CTF), as shown in (Figure 4), demonstrated the crawler's ability to navigate complex geometries and remove debris effectively. Feedback from stakeholders has led to plans for further refinement of the system, including improvements to communication and power circuits and the construction of a more robust metal chassis.



Figure 4. Cold Test Facility

This report provides an in-depth analysis of the project's methodologies, testing procedures, and outcomes, highlighting the potential of the vacuum crawler system to revolutionize nuclear facility maintenance and hazardous waste management. The ongoing development efforts aim to enhance the system's reliability, effectiveness, and safety, in alignment with the broader mission of advancing the U.S. Department of Energy’s environmental management initiatives.

2. RESEARCH DESCRIPTION

2.1 Deployment strategies

The initial stage of the internship focused on developing a deployment method for the crawler through the Mockup Riser at the Cold Test Facility (Figure 5). Key considerations included maintaining a controlled descent speed and ensuring the crawler remains parallel to the floor to prevent the front nozzle from crashing into the floor. This was achieved by integrating an additional suspension cable, secured within the umbilical cord sleeve, to prevent entanglement with the main safety suspension cable. This design ensures that the crawler can be manually retrieved in the event of a mechanical failure.

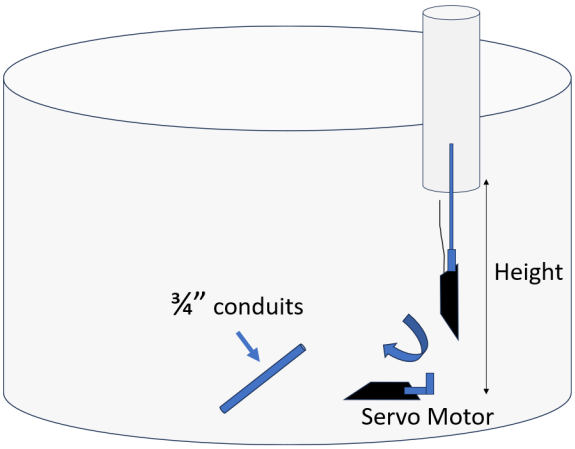


Figure 5. Deployment Strategy



Figure 6. Cold Test Facility Double-Shell Mockup

A structure was designed to mount on the flange of the mockup riser, providing easy access to hoist controls in case of entanglement during deployment (Figures 7-9).

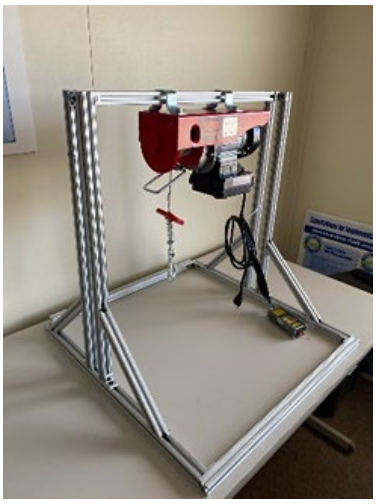


Figure 7. Completion of Manufactured Deployment Mechanism



Figure 8. Assembly of Deployment Mechanism



Figure 9. Final Deployment Mechanism Installation on Mockup Riser at the Cold Test Facility

The second stage involved manufacturing the main harness, which consists of a series of air hoses stepped down in size from 3/4” to 1/2” to 3/8”. This configuration reduces the airflow which results in a reduction of the air pressure from 120 psig at the compressor to 90 psig at both inlets of the vacuum line modules, while also making the air circuit more manageable for manual deployment. The harness includes a vacuum hose connected to a replaceable debris collector suspended 15 feet above the crawler, allowing easy access for bag replacement during cleaning operations.

The correct lengths for the harness assembly were determined for each section (Figure 10), which consists of three parts: the deployment end, the bag chamber, and the main crawler connection end. The main connection includes the vacuum hose, electrical cable, leveling cable, and safety cable. To prevent tangling, each section’s length was precisely measured and assembled.

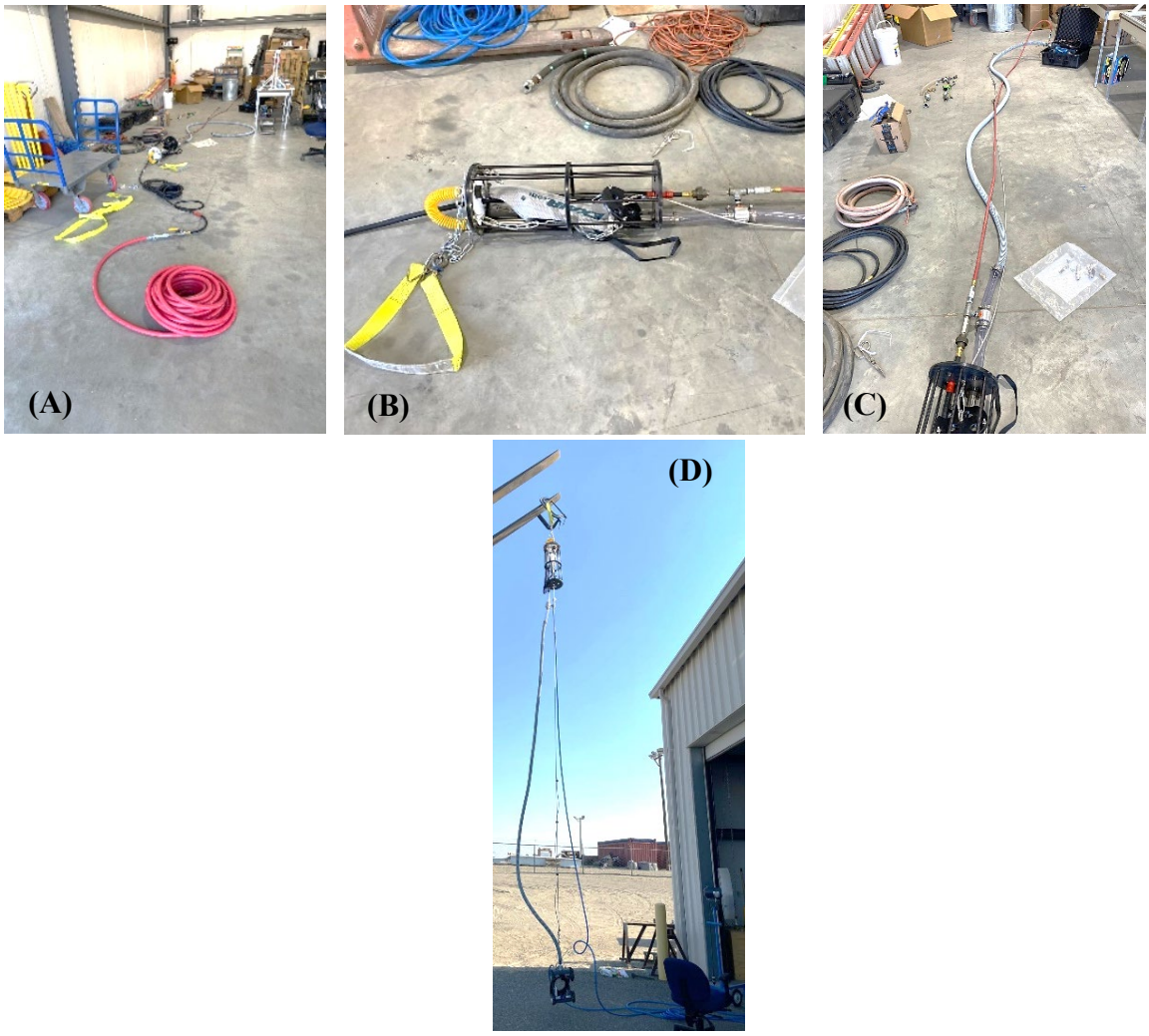


Figure 10. Assembly Process of Main Harness. (A): Compressed air line, (B): Vacuum bag assembly, (C): Vacuum bag assembly with compressed air and vacuum hoses, (D): Complete assembly suspended from a forklift with the rover attached on the bottom.

The third stage focused on troubleshooting different airline circuits to optimize the performance of the vacuum components (Figures 11-12).



Figure 11. Air Compressor and Air Lines



Figure 12. Demonstration Configuration

2.2 Design

The VCS is composed of several critical components, engineered to navigate and clean the annulus space within DSTs at the Hanford site. The primary elements of the design include:

Pneumatic Vacuum Module:
The crawler is equipped with two Exair Line Vac pneumatic vacuum modules, positioned in series to optimize suction power. One module is mounted directly on the crawler, while the other is integrated further along the umbilical harness. This configuration ensures maximum airflow and pull force at the cleaning nozzle, enhancing the crawler's debris collection capabilities.
An appropriately sized portable air compressor, rated at 185 scfm, provides the necessary flow rate to operate the vacuum module.

Crawler Chassis and Mobility System:

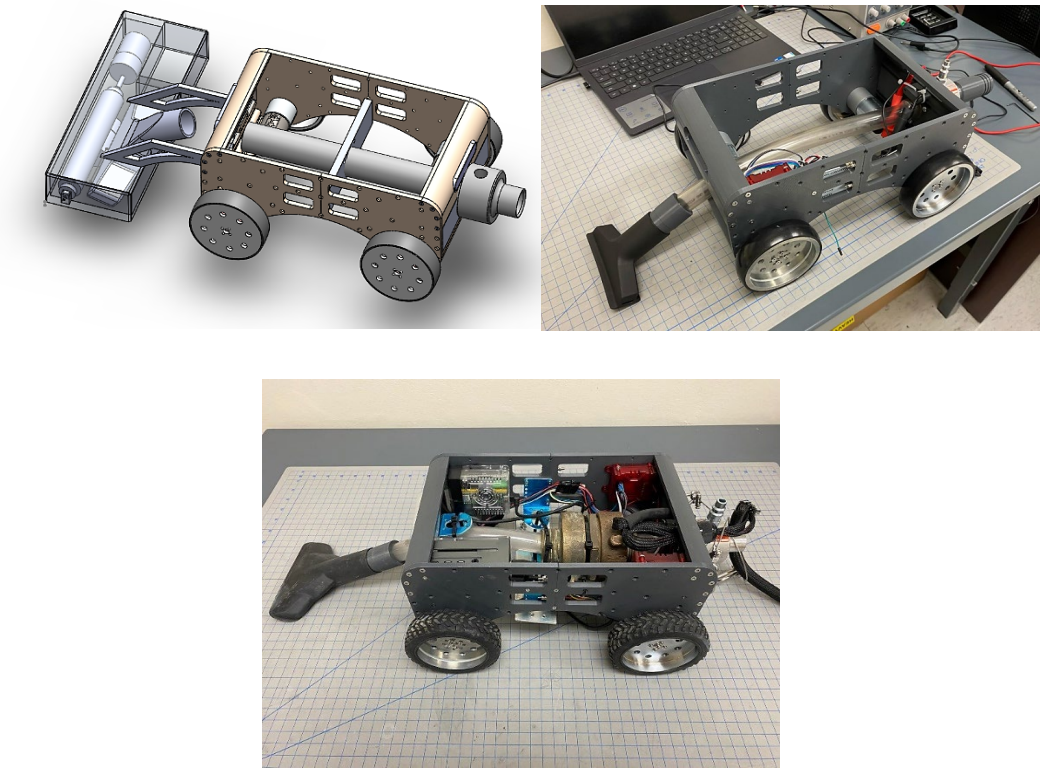


Figure 13. Chassis Prototype

The prototype chassis (Figure 13) is constructed from acrylonitrile butadiene styrene (ABS) 3D-printed plastic. However, future iterations will incorporate materials such as stainless steel, anodized aluminum, or non-porous plastics to ensure compatibility with the nuclear environment and meet radiation tolerance requirements.

The crawler employs a differential drive system, utilizing four independently controlled wheels to navigate the 30-inch-wide annulus space. The system is designed to traverse obstacles such as welds and conduits up to 1/2 inch in height, with rubber tread nubs on the wheels providing enhanced grip and stability.

Control and Feedback Systems:



Figure 14. Control Room

The system's control features will consist of bidirectional brush selectors, variable light controls, and an intuitive joystick interface. Real-time video feedback with low latency is provided through onboard cameras, allowing operators to control the crawler from a distance of up to 300 feet (Figures 14-15).



Figure 15. 300ft Power/Communication CAT 7-Reel Cable

The system operates without the need for software or firmware, relying instead on analog control circuits to manage motor functions, video feedback, and other sensor data. This analog control circuit approach minimizes potential points of failure and simplifies the system's operation in the field.

Debris Collection and Surface Agitation:
The vacuum module is designed to work in conjunction with mechanical assemblies that will collect debris from the DST and reduce particle sizes to 1/2 inch or less for more efficient vacuum removal (Figure 16). The system will also incorporate multiple cleaning stages to further enhance vacuum efficiency.

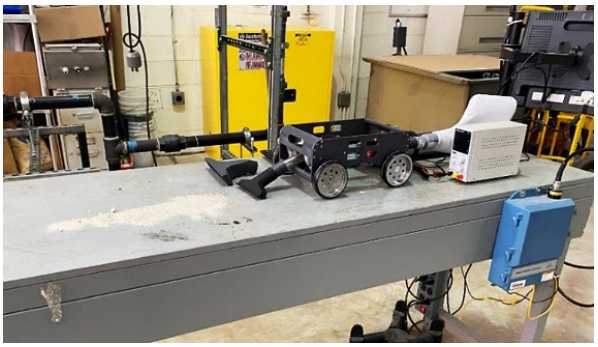



Figure 16. Debris Collection Test

A multistage mechanical system is planned for future development, enabling the crawler to manipulate and remove foreign objects that cannot be vacuumed. This system deposit objects into a secure container designed to prevent spillage during transport.

2.3 Testing Procedures

The VCS underwent rigorous testing at CTF to evaluate its performance under conditions simulating the DST environment. Key testing procedures included:

Vacuum Performance Testing:
The suction power and airflow of the Exair Line Vac modules were tested using various aggregates of debris and rocks. Tests were conducted on both 1" and 1-½" vacuum lines to validate the system's ability to handle different debris sizes and types. The system's ability to maintain suction and prevent clogging was assessed by running the vacuum module continuously while navigating the mock annulus.

Maneuverability Testing:
The crawler's ability to traverse obstacles and navigate around welds up to ½ inch in height was tested. The system's design and ground clearance were evaluated to ensure reliable performance on warped liner bottoms and slopes within the annulus (Figure 17). Differential drive control systems were fine-tuned to optimize the crawler's responsiveness and stability during operation.



Figure 17. Control Room Camera Feed

Control System Validation:
The analog control circuits were tested and validated to ensure reliable operation without dependencies on software or firmware. The joystick interface was programmed to control all motors, and the system's response time to operator inputs was measured to be within 1 second, using a manual timer and adjustments to the joystick calibration.

3. RESULTS AND ANALYSIS

The VCS underwent a series of tests at CTF to evaluate its performance in simulating the conditions within the annulus space of DSTs. The key areas tested included suction power, maneuverability, debris removal efficiency, and control system responsiveness.

Vacuum Module Configuration:
The system used dual EXAIR Line Vac pneumatic vacuum modules arranged in series. These modules, connected to standard hoses or tubes, act as in-line conveyors, efficiently moving large volumes of material over long distances. Compressed air is injected through nozzles, creating a vacuum that generates high output flows. The design features a large, smooth bore to maximize material throughput, and its maintenance-free operation is ensured by the absence of moving parts.

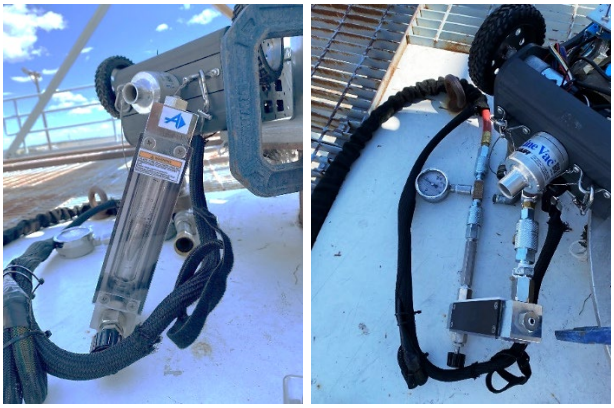


Figure 18. Collecting air flow data

The conveying rate is influenced by various factors, including the size and mass of the material, as well as the length, lift, and bends in the hose or tube. Hands-on testing is crucial for optimizing configurations, such as selecting appropriate fittings, to ensure efficient flow rates (Figure 18). The data collected during these trials are shown in Table 1.

Table 1. Vacuum Performance and Debris Removal Efficiency

Media	Test	Line Vac Diameter	Vertical Rise (ft)	Rate (scfm)	Aggregate Debris (g)	Debris Removed (g)	Percentage Removed (%)
Aggregates of debris and rocks (Density: 68 lbs/ft^3)	1	6081 (1” Aluminum)	15	4	500	385	77.0%
	2	6083 (1 ½” Aluminum)	30	2	500	110	22.0%
	Vac lines in Series						
	3	1 + 1 ½”	15	7	500	475	95.0%
	4	1 + 1 ½”	30	6	500	425	85.0%

Optimal performance of the EXAIR Line Vac requires maintaining a steady supply of compressed air at the correct pressure. Factors such as compressor output pressure, airflow rate, piping diameter, and internal pipe smoothness can significantly affect performance. The air compressor must provide both sufficient pressure and flow; inadequate flow can impair performance even if

the pressure is sufficient. A Sullair air compressor, capable of supplying up to 185 scfm at 120 psig, was used (Figure 19).



Figure 19. Sullair Air Compressor

We have observed that operating below 80 psig can reduce performance, while operating above this pressure may provide only marginal gains but at the cost of increased energy consumption. Proper selection of connectors and fittings is crucial to avoid restricting airflow, as restrictions can significantly impact tool performance. Quick connectors with smaller diameters can diminish airflow, so selecting the correct pipe size is essential to minimize pressure loss over distance.

The ability to convey abrasive media was crucial to system success. The Line Vac's no-moving-parts design ensures reliability and maintenance-free operation, as demonstrated in Figure 20.

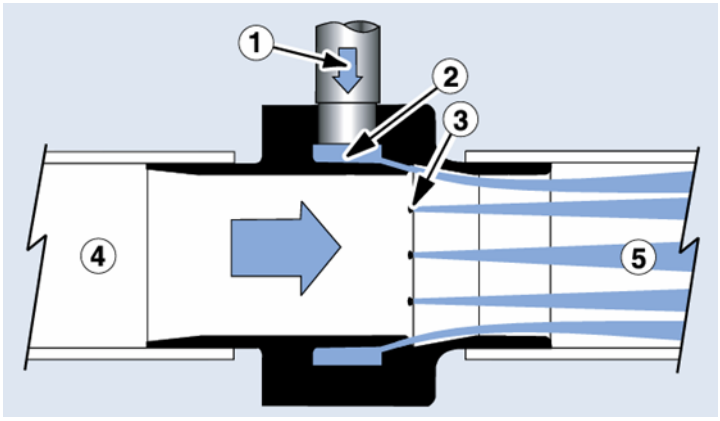


Figure 20. Line Vac

Figure 20 shows how compressed air flows through the inlet (1) into an annular plenum chamber (2). It is then injected into the throat through directed nozzles (3), creating a vacuum at the intake (4), which draws in material and accelerates it through the unit (5) for conveyance over long vertical or horizontal distances.

3.1 Analysis

The test results indicate that the VCS demonstrates strong performance across several key metrics, particularly in terms of vacuum efficiency and maneuverability.

Vacuum Efficiency:
The EXAIR Line Vac pneumatic vacuum modules proved highly effective in removing debris from the annulus space. As shown in Table 1, debris removal percentages ranged from 22.0% to 95.0%, with the highest efficiency achieved using a combination of 1-inch and 1 ½-inch vacuum lines at 15 feet. The increased diameter likely enhances airflow and reduces the risk of clogging, offering valuable insights for future design optimizations.

Maneuverability:
The crawler successfully navigated the required obstacle height of 1/2 inch, validating the robustness of the differential drive system. Maintaining consistent speed while overcoming obstacles is crucial for reliable operation in the challenging annulus environment, as detailed in Table 2. Future tests will be necessary to account for the additional weight from the metal chassis redesign. Additionally, the crawler executed a full 360-degree turn within the 30-inch gap and incline of the shell.

Table 2. Maneuverability and Obstacle Traversed Ability

Test	Crawlers Weight (lbs)	Obstacle Height (in)	Travel Time in 2ft (sec)	Obstacle Traversed (Y/N)
1	10	0	8	Y
2	10	⅜	12	Y
3	15	½	22	Y
4	15	¾	27	Y
5	10	1 ½	NA	N

Control System Performance:
The control system, featuring a joystick interface and analog circuits, delivered excellent precision. This precision is crucial for maintaining the stability of the crawler and ensuring smooth, reliable operation in the confined and complex environment of the annulus space. The system's responsiveness and accuracy instill confidence in the crawler's ability to navigate tight spaces and perform tasks with minimal operator intervention.

Overall, the VCS's performance in these tests is promising, indicating that the current design meets the fundamental requirements for DST annulus cleaning (Figures 21-22). The insights gained from

this comprehensive analysis will guide further refinement of the system, focusing on optimizing vacuum efficiency and ensuring the crawler's durability and reliability in the nuclear environment.



Figure 18. Deployment Day Demo



Figure 19. Successful Deployment Positioning

4. CONCLUSION

The development and testing of the vacuum crawler system successfully demonstrated its capability to address the critical challenges associated with the inspection and cleaning of the annulus space in Double-Shell Tanks (DSTs) at the DOE's Hanford site (Figure 23). The crawler effectively utilized the dual Exair Line Vac pneumatic vacuum modules to achieve high debris removal efficiency, while the robust design ensured reliable maneuverability over obstacles up to ½ inch in height. The control system provided precise operation, with low-latency video feedback and accurate joystick control contributing to the overall effectiveness of the system. The crawler effectively performed its primary function of debris removal, achieving removal rates of up to 95% under various testing conditions, thus validating its design and operational capabilities.



Figure 20. Demonstration

The success of the initial testing trials indicates that the vacuum crawler system (VCS) is well-positioned to advance into further stages of development, with the potential to significantly enhance the safety and efficiency of hazardous waste management in nuclear facilities. Insights gained from these trials will guide the next steps in refining the system's design. Planned upgrades include replacing the prototype chassis with a more robust design using stainless steel or anodized aluminum to withstand the harsh conditions within double-shell tanks (DSTs). Further development will improve vacuum efficiency, focusing on enhancing debris collection with upgrades to the vacuum module and brush assemblies. The multistage mechanical system will be refined to ensure reliable retrieval and containment of foreign objects, along with enhancements to the brush assembly and debris container. Additionally, the control system will be upgraded with added sensor data feedback, such as temperature, pressure, and flow, providing operators with more comprehensive monitoring capabilities. These planned enhancements aim to optimize the vacuum system and integrate additional mechanical assemblies for surface agitation, contributing to a more effective and reliable hazardous waste management solution.

This project demonstrated the viability of using remotely operated robotic systems for complex environmental management tasks, contributing to the broader mission of the U.S. Department of Energy's environmental remediation efforts. With continued development, the VCS has the capability to become a valuable tool in supporting the maintenance of nuclear storage facilities.

5. REFERENCES

[1] RPP-PLAN-43988 Rev.08A, "Technology and Innovation Roadmap, Rev. 8a," Washington River Protection Solutions, LLC, Richland, Washington, 2023.

[2] Department of Energy. Accessed on August 20, 2024. Site: https://www.energy.gov/

6. APPENDIX

Technology Element Description Summary (TEDS) MTW-89:

RPP-PLAN-43988
Revision 8a

Technology and Innovation Roadmap

Prepared by

A.L.Hede
Washington River Protection Solutions, LLC

Date Published
November 2023

washington river protection solutions

Prepared for the U.S. Department of Energy
Office of River Protection

Contract No. DE-AC27-08RV14800

Approved for Public Release;
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A-6007-231 (REV 0)



RPP-PLAN-43988, Rev. 8a

CHIEF TECHNOLOGY OFFICE

washington river protection solutions

HANFORD SITE
US DEPT OF ENERGY

Remote Concrete Surface Cleaning Apparatus

TEDS ID: MTW-89 **Timetable: > 5 Years**

A remotely operated cleaning tool for concrete walls, waste transfer pits, ceiling cover blocks that are required to be painted with a SPC. This device must be able to thoroughly clean the SPC (i.e., Amerlock® 2/400 resin) without damaging it.

TECHNOLOGY NEED

A technology is needed to allow for remote cleaning of concrete walls, waste transfer pits and ceiling cover blocks that are painted with a special protective coating (SPC). Per WAC-173-303, SPC is required for concrete surfaces that may come in contact with tank waste (e.g., tanks vaults process pits, valve pits) must be inspected and repaired as necessary to maintain RCRA permit compliance. These surfaces will get dirty and contaminated due to occasional spills during operations. Before proper inspection and repair can occur the surfaces must be cleaned. Due to high radiation fields noted in most pits, the work associated with the cleaning and required coating repairs must be completed remotely (through the use of extension poles). Inspection of the cleaned and repaired surfaces is completed by visual inspection of high quality digital photographs.

TECHNOLOGY SOLUTION

The DST Annulus Floor Cleaning System, developed by Rolls-Royce for use by WRPS, is a remote operated cleaning tool designed to vacuum debris in DSTs annulus. It is believed that this technology could be further modified to be used in transfer pits and other RCRA facility locations to perform the necessary cleaning.



Typical Transfer Pit Configuration, Areas that Need to be Cleaned



Robotic Cleaner with Vacuum

Technology Maturation Level

Research and Concept

National Laboratory Involvement?

No

Rough Order of Magnitude Cost & Duration?

$0 - $1 Million

0 Years - 2 Years

THREATS AND OPPORTUNITIES

TFIRR-0054-T: Pit Corrosion
DAORR-0017-T: Pit Corrosion
DSTI23-0001-T: DST Failure

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